PREVUMO

PREVentive Usage MOnitor



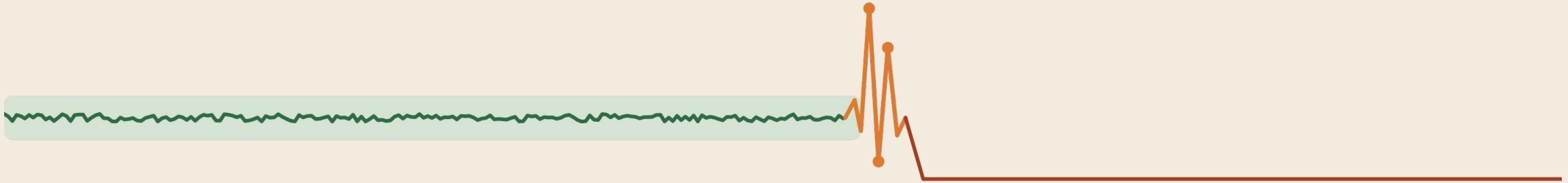
Know what happened to your machine, and when it's about to fail.

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GNG4120 · Group 10 · Oct 23, 2026

A $3,000 welder dies on a Tuesday.

Nobody saw it coming.



The owner knows who booked it. Not what happened to it.

Machines break, and nobody sees it coming

More makerspaces, labs and shared kitchens now rent out expensive electric equipment.

$754

Average emergency repair call

“Hundreds if not thousands of dollars” per breakdown, a makerspace director told us

$600 to $6,000

What each lost member is worth

When machines keep breaking, members cancel

No history

No sensor sees the warning signs

Booking tools only work “when people actually use the software”

Sources: makerspace director interview (Group 10, 2026); emergency repair call average, 86 Repairs 2024, via Envigilance.

Who we serve, and what we promise

Owners and managers of shared spaces with costly electric equipment

- Makerspaces
- University labs
- Shared kitchens

One owner can run several locations.

Value proposition

Catch failures before they happen,

know who used each machine and how much,

and shut it down safely from anywhere.

How it works

1



Box on the machine

Keypad, current sensor, relay. ESP32 with Wi-Fi.

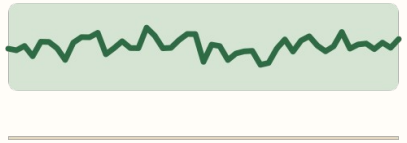
2



Wi-Fi to the cloud

Every second of current, per session.

3



Learns its curve

Each machine gets its own normal range.

4

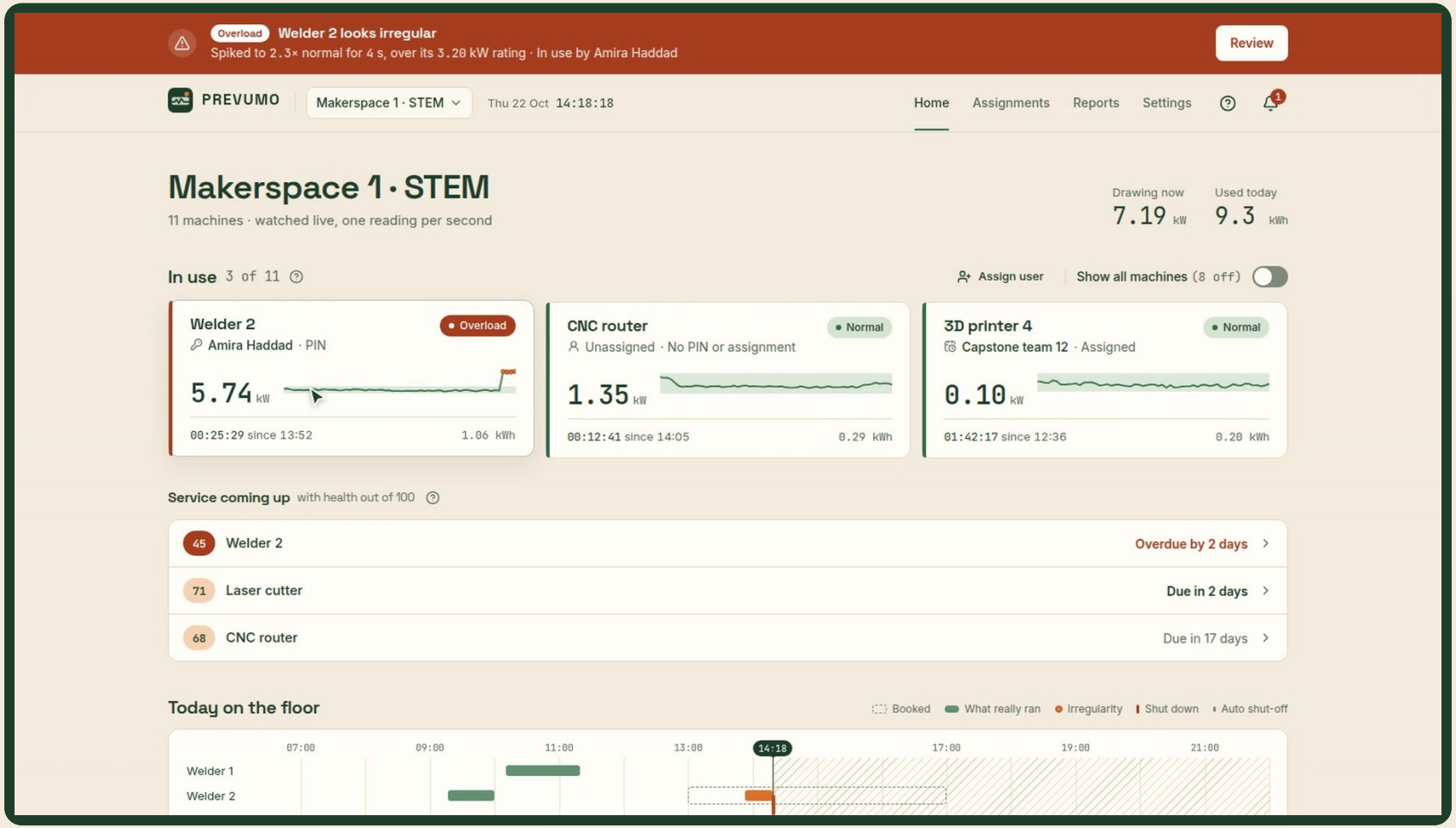


Owner decides

Alert, then one tap.

Safety: no auto restart after a shutdown, and fridges are never cut.

The owner app, live



Owner side prototype. Simulated boxes, real flow. Click to play.

In the video

Login, alert, technician, idle shutdown, assign, daily report.

A market that runs on electric equipment

31

shared-use commercial kitchens in Ontario

poidata.io, Aug 2025

30 to 40%

saved by predictive maintenance versus fixing after breakdown

U.S. DOE, O&M Best Practices Guide

5 to 7×

more energy per sq ft in restaurants than other commercial buildings

ENERGY STAR



Expansion path

One makerspace → University networks with several labs → Shared kitchens

What each one knows

Product	Category	Price	What it knows
Skedda	Booking	US$249+ /mo	Who booked
Fabman	Machine access	€49 to €199 /mo + €10 to €14 per machine	Who switched it on
Emporia	Energy monitor	~US$200	How much energy
MachineMetrics	Factory monitoring	~US$1,400+ hardware per kit	Machine health, for factories only
PREVUMO	Shared spaces	Assumption, slide 9	**Who used it + how much + machine health + remote shutdown**

Skedda could be a partner, not just a competitor: a booking becomes a PIN on our box.

Prices from each company's public pricing, gathered in our competitor research.

From billing to predictive maintenance

Plan A (PD A)

Usage tracker and billing

Partners	Activities	Value	Relations	Segments
Component suppliers, electricians	Box + app, billing reports	Bill each user for energy	Self-service app	Coworking, makerspaces, kitchens, labs
	Resources: Usage data per user		Channels: Website, outreach	
Costs: Box, install, hosting		Revenue: ~$150 box + ~$10/machine/mo		

Plan B (now)

Safety and predictive maintenance

Partners	Activities	Value	Relations	Segments
Partners: Suppliers, **technician network**, booking platforms	**Activities**: Box + app, **AI curve learning, alerts**	**Value**: **Warn before failure**, remote shutdown, idle shut-off	Relations: App + **alert follow-up**	**Segments**: **Makerspaces, labs** first, then kitchens
	Resources: **Anomaly model, irregularity history**		Channels: Website, outreach, **partner referrals**	
Costs: Box, install, hosting + **AI development**		**Revenue**: ~$300 box + ~$25/machine/mo + referral		

Orange = changed

Why it changed: Every owner we asked bundles electricity into rent or membership, so per-user billing isn't the pain. Breakdowns and equipment left on are.

Revenue (assumptions)

$300
box per machine, one time

$25
per machine per month

+ possible referral fee on technician jobs

Per machine, year 1
$300 box + $300 subscription
= $600 revenue
Assumptions to validate

What we've tested so far

6
competitors reverse engineered

2
business model canvases

49
outreach emails, Canada, US, Europe
replies still coming in

10
interviews

The pivot
Billing → predictive maintenance

> **"Managing equipment is one of the biggest tasks we have."**
>
> Executive Director, Ontario makerspace

> **"I pay all the electricity and build that price into my monthly rent. Simple, easy, fair."**
>
> Owner, Manitoba shared kitchen

Assumption	Method	Result
Owners bill electricity per user	Interviews	**Invalidated**
Owners want warning before failures	Interviews	**Partly supported**
Staff switch equipment off at closing	Café interviews	**Invalidated: control must sit with the owner**

Full validation board in the appendix.

Every owner we asked builds electricity into rent or membership, so the headline moved from billing to predictive maintenance. And café staff forget equipment on, so we added scheduled shutdowns.

Skedda tells you who booked it.

PREVUMO tells you who used it, and when it’s about to fail.



Thank you. Questions?

Remote shutdown hardware

Up to 15 A at 120 V

A certified relay inside the box switches the machine directly.

240 V or 3-phase

The box drives a contactor, installed by a licensed electrician.

Before we sell

CSA certification of the box is required.

After any shutdown the machine stays off until someone starts it again. Fridges and freezers are never cut.

Learning, privacy and limits

About 7 days

to learn a new machine's normal current pattern. Alerts switch on after that.

Owner only

Usage data belongs to the owner's account. Members don't see each other's sessions.

Electric only

Gas equipment is invisible to electricity data.



Validation board

Customer	Problem	Core assumption	Method	Success criterion	Result after this week
Makerspace and kitchen owners	Energy cost per user is invisible	Owners want to bill electricity per user	Interviews: 3 owners, Magna, IEEE	Owners charge members for energy	**Invalidated: 3 of 3 owners bundle it into rent or fees**
Makerspace, manufacturing, café	Breakdowns cost money and time	Breakdowns are a costly, frequent pain	Interviews	6 of 10 describe a costly breakdown	**Partly supported: makerspace and Magna give costs, 3 café staff; 6 of 10 not met yet**
Makerspace and lab managers	Failures come without warning	Owners want warning before a failure	Interviews	Failures are noticed too late	**Partly supported: café staff notice only once it stops**
Cafés and kitchens	Equipment left on after closing	Owners need idle and scheduled shutdown	Café interviews, manager field notes	Equipment found on after closing	**Supported, new: something left on almost every night**
Café staff	Nobody owns switching off	Staff switch equipment off at closing	Café interviews	Staff switch off reliably	**Invalidated: control must sit with the owner**
University student spaces	Who used which machine	Open-access spaces need per-machine tracking	IEEE uOttawa interview	They track use per machine	**Invalidated: RFID sign-in only, free equipment not tracked**

Q&A prep

Why not a smart plug?

No PIN, no per member history, no learned normal curve, and most can't carry a welder.

Heavy 240 V equipment?

The box drives a contactor installed by a licensed electrician.

Liability for a shutdown?

The owner decides, every shutdown is logged, nothing restarts on its own.

How does it know normal?

About 7 days of the machine's own current, then it flags sustained changes.

Can a non-profit pay $25?

One avoided emergency call ($754) covers about 2.5 machine-years. Price still to validate.

Versus Fabman?

Fabman knows who switched it on. We add machine health, idle and technicians.

What did interviews change?

Every owner we asked bundles electricity into rent, so we dropped per-user billing as the headline and made predictive maintenance the core.

Full answers in the speaker notes.

Sources

- Group 10 interviews, Oct 21 to 23, 2026 (makerspace, two shared kitchens, manufacturing, café staff, university student office). Quotes anonymized.
- Emergency repair call average, $754: 86 Repairs, 2024, via Envigilance.
- 31 shared-use commercial kitchens in Ontario: poidata.io, August 2025.
- Predictive maintenance saves about 30 to 40% versus reactive maintenance: U.S. Department of Energy, Operations & Maintenance Best Practices Guide.
- Restaurants use 5 to 7 times more energy per square foot than other commercial buildings: ENERGY STAR.
- Competitor prices: Skedda, Fabman, Emporia and MachineMetrics public pricing, from our competitor research.
- Prices for PREVUMO ($300 box, $25 per machine per month) are assumptions to validate.